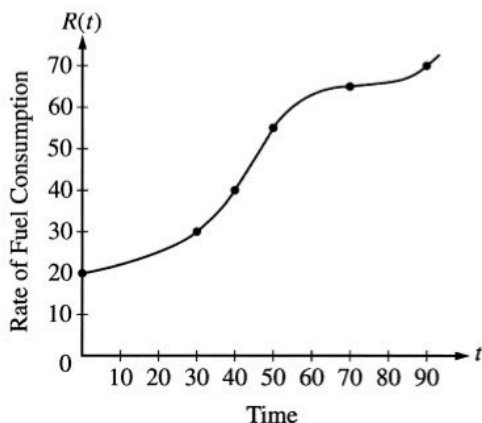


6-2b

1.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

Approximate the value of $\int_0^{90} R(t) dt$ using a left Riemann sum with the five subintervals indicated by the data in the table. Is this numerical approximation less than the value of $\int_0^{90} R(t) dt$? Explain your reasoning.

Part C

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.

6-2b

t (minutes)	0	2	5	7	11	12
$r'(t)$ (feet per minute)	5.7	4.0	2.0	1.2	0.6	0.5

The volume of a spherical hot air balloon expands as the air inside the balloon is heated. The radius of the balloon, in feet, is modeled by a twice-differentiable function r of time t , where t is measured in minutes. For $0 < t < 12$, the graph of r is concave down. The table above gives selected values of the rate of change, $r'(t)$, of radius of the balloon over the time interval $0 \leq t \leq 12$. The radius of the balloon is 30 feet when $t = 5$.

(Note: The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.)

2. Is your approximation in part (c) greater than or less than $\int_0^{12} r'(t) dt$? Give a reason for your answer.

Part D

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$

Unit Point

1 point is earned for: units in (b) and (c)

Units of ft³/min in part (b) and ft in part (c)

6-2b



0	1
---	---

The student response earns all of the following points:

1 point is earned for: units in (b) and (c)

Units of ft³/min in part (b) and ft in part (c)

3. Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_0^{12} r'(t) dt$.
Using correct units, explain the meaning of $\int_0^{12} r'(t) dt$ in terms of the radius of the balloon.

Part C

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5) \\ = 19.3 \text{ ft}$$

$$\int_0^{12} r(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5) \\ = 19.3 \text{ ft}$$

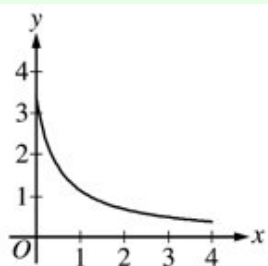
$$\int_0^{12} r(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

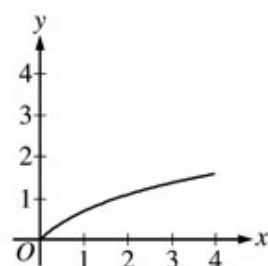
6-2b

4. If a trapezoidal sum overapproximates $\int_0^4 f(x)dx$, and a right Riemann sum underapproximates $\int_0^4 f(x)dx$, which of the following could be the graph of $y = f(x)$?

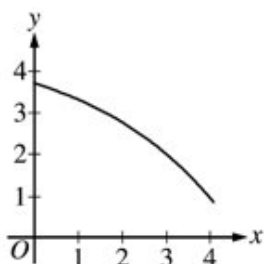
(A)



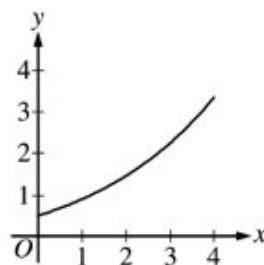
(B)



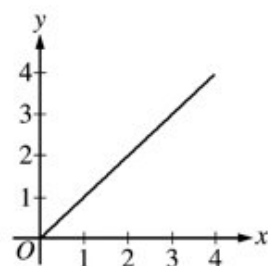
(C)



(D)



(E)



6-2b

5.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

(a) Use the data in the table to approximate Ben's acceleration at time $t = 5$ seconds. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{60} |v(t)| dt$ in the context of this problem. Approximate

$\int_0^{60} |v(t)| dt$ using a left Riemann sum with the subintervals indicated by the data in the table.

(c) For $40 \leq t \leq 60$, must there be a time t when Ben's velocity is 2 meters per second? Justify your answer.

(d) A light is directly above the western end of the track. Ben rides so that at time t , the distance $L(t)$ between Ben and the light satisfies $(L(t))^2 = 12^2 + (B(t))^2$. At what rate is the distance between Ben and the light changing at time $t = 40$?

Part A

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)

Part B

1 point is earned for meaning of the integral

6-2b

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$$\int_0^{60} |v(t)| dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139 \text{ meters}$$

1 point is earned for units in (a) or (b)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$$\int_0^{60} |v(t)| dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139 \text{ meters}$$

1 point is earned for units in (a) or (b)

Part C

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\frac{B(60) - B(40)}{60 - 40} = \frac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t , $40 < t < 60$, such that $v(t) = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for difference quotient

6-2b

1 point is earned for conclusion with justification

Because $\frac{B(60) - B(40)}{60 - 40} = \frac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t , $40 < t < 60$, such that $v(t) = 2$.

Part D

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

6-2b

6.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

Using correct units, interpret the meaning of $\int_0^{60} |v(t)| dt$ in the context of this problem. Approximate $\int_0^{60} |v(t)| dt$ using a left Riemann sum with the sub intervals indicated by the data in the table.

Part B

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t=0$ to $t=60$.

$\int_0^{60} |v(t)| dt \approx 2.0 * 10 + 2.3(40-10) + 2.5(60-40) = 139$ meters



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t=0$ to $t=60$.

$\int_0^{60} |v(t)| dt \approx 2.0 * 10 + 2.3(40-10) + 2.5(60-40) = 139$ meters

6-2b

7.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	-20	-30	-20	-14	-10	0	10
$a(t)$ (ft/sec ²)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \leq t \leq 60$ seconds, the car's velocity v , measured in feet per second, and acceleration a , measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

(a) Using appropriate units, explain the meaning of $\int_{30}^{60} |v(t)| dt$ in terms of the car's motion. Approximate

$\int_{30}^{60} |v(t)| dt$ using a trapezoidal approximation with the three subintervals determined by the table.

(b) Using appropriate units, explain the meaning of $\int_0^{30} a(t) dt$ in terms of the car's motion. Find the exact value of $\int_0^{30} a(t) dt$.

(c) For $0 < t < 60$, must there be a time t when $v(t) = -5$? Justify your answer.

(d) For $0 < t < 60$, must there be a time t when $a(t) = 0$? Justify your answer.

Part A

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)



0	1	2	3
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6-2b

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part B

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t) dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_0^{30} a(t) dt = \int_0^{30} v'(t) dt = v(30) - v(0)$$

$$= -14 - (-20) = 6 \text{ ft/sec}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

6-2b

1 point is earned for the value

$\int_0^{30} a(t) dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_0^{30} a(t) dt = \int_0^{30} v'(t) dt = v(30) - v(0)$$

$$= -14 - (-20) = 6 \text{ ft/sec}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part C

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

Part D

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0, 25)$ so that $a(t) = v'(t) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

6-2b

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t) = v'(t) = 0$.

6-2b

8.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	-20	-30	-20	-14	-10	0	10
$a(t)$ (ft/sec ²)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \leq t \leq 60$ seconds, the car's velocity v , measured in feet per second, and acceleration a , measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

Using appropriate units, explain the meaning of $\int_{30}^{60} |v(t)| dt$ in terms of the car's motion. Approximate

$\int_{30}^{60} |v(t)| dt$ using a trapezoidal approximation with the three sub intervals determined by the table.

Part A

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t=30$ sec to $t=60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 ft$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t=30$ sec to $t=60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 ft$$

6-2b

9.

t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

Concert tickets went on sale at noon ($t = 0$) and were sold out within 9 hours. The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Values of $L(t)$ at various times t are shown in the table above.



Use a trapezoidal sum with three subintervals to estimate the average number of people waiting in line during the first 4 hours that tickets were on sale.

Part B

The response can earn up to 2 points:

1 point: For correct use of trapezoidal sum

1 point: For correct answer

The average number of people waiting in line during the first 4 hours is

approximately

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right) =$$

155.25 people



0	1	2
---	---	---

The response earns both of the following points:

1 point: For correct use of trapezoidal sum

1 point: For correct answer

The average number of people waiting in line during the first 4 hours is

approximately

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right) =$$

155.25 people

6-2b

10.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

Write an integral expression in terms of $T(x)$ for the average temperature of the wire. Estimate the average temperature of the wire using a trapezoidal sum with the four subintervals indicated by the data in the table. Indicate units of measure.

Part B

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\frac{1}{8} \int_0^8 T(x) dx$:

$$A = \frac{100+93}{2} \cdot 1 + \frac{93+70}{2} \cdot 4 + \frac{70+62}{2} \cdot 1 + \frac{62+55}{2} \cdot 2$$

$$\text{Average temperature} \approx \frac{1}{8} A = 75.6875^{\circ}\text{C}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\frac{1}{8} \int_0^8 T(x) dx$:

$$A = \frac{100+93}{2} \cdot 1 + \frac{93+70}{2} \cdot 4 + \frac{70+62}{2} \cdot 1 + \frac{62+55}{2} \cdot 2$$

6-2b

Average temperature $\approx \frac{1}{8}A = 75.6875^\circ C$

6-2b

11.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t=0$) and 8 P.M. ($t=8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

Use a trapezoidal sum with the four subintervals given by the table to approximate the value of $\frac{1}{8} \int_0^8 E(t) dt$.

Using correct units, explain the meaning of $\frac{1}{8} \int_0^8 E(t) dt$ in terms of the number of entries.

Part B

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

$$\begin{aligned} \frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688 \\ \frac{1}{8} \int_0^8 E(t) dt \end{aligned}$$

is the average number of hundreds of entries in the box between noon and 8 P.M



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

6-2b

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688 \\ \frac{1}{8} \int_0^8 E(t) dt\end{aligned}$$

is the average number of hundreds of entries in the box between noon and 8 P.M

6-2b

12.

t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

Approximate the average temperature, in degree Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using a trapezoidal approximation with subintervals of length $\Delta t = 3$ days.

Part B

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^\circ\text{C}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^\circ\text{C}$$

6-2b

13.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

Using correct units, explain the meaning of the definite integral $\int_0^{40} |v(t)| dt$ in the context of the problem.

Approximate the value of $\int_0^{40} |v(t)| dt$ using a right Riemann sum with the four subintervals indicated in the table.

Part B

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$

$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$

$= 2400 + 1920 + 880 + 2400$

$= 7600$ meters



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$

6-2b

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$

6-2b

14.

t (seconds)	0	8	20	25	32	40
$v(t)$ (meters per second)	3	5	-10	-8	-4	7

The velocity of a particle moving along the x -axis is modeled by a differentiable function v , where the position x is measured in meters, and time t is measured in seconds. Selected values of $v(t)$ are given in the table above. The particle is at position $x=7$ meters when $t=0$ seconds.

Using correct units, explain the meaning of $\int_{20}^{40} v(t) dt$ in the context of this problem. Use a trapezoidal sum with the three subintervals indicated by the data in the table to approximate $\int_{20}^{40} v(t) dt$.

Part B

1 point is earned units in (a) and (b)

1 point is earned for meaning of $\int_{20}^{40} v(t) dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t) dt$ is the particle's change in position in meters from time $t=20$ seconds to time $t=40$ seconds.

$$\int_{20}^{40} v(t) dt = \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$

= -75 meters



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned units in (a) and (b)

1 point is earned for meaning of $\int_{20}^{40} v(t) dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t) dt$ is the particle's change in position in meters from time $t=20$ seconds to time $t=40$ seconds.

$$\int_{20}^{40} v(t) dt = \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$

= -75 meters

6-2b

6-2b

15.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	-120	-150

Train A runs back and forth on an east-west section of railroad track. Train A 's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

At time $t = 2$, train A 's position is 300 meters east of the Origin Station, and the train is moving to the east. Write an expression involving an integral that gives the position of train A , in meters from the Origin Station, at time $t = 12$. Use a trapezoidal sum with three subintervals indicated by the table to approximate the position of the train at time $t = 12$.

Part C

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$s^A(12) = s^A(2) + \int_2^{12} V_A(t) dt = 300 + \int_2^{12} V_A(t) dt$$

$$\int_2^{12} V_A(t) dt \approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2}$$

$$S_A(12) \approx 300 - 450 = -150$$

The position of Train A at time $t=12$ minutes is approximately 150 meters west of Origin Station.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$s^A(12) = s^A(2) + \int_2^{12} V_A(t) dt = 300 + \int_2^{12} V_A(t) dt$$

$$\int_2^{12} V_A(t) dt \approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2}$$

6-2b

$$= -450$$

$$S_A(12) \approx 300 - 450 = -150$$

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.

6-2b

16.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

Using correct units, explain the meaning of $\int_{10}^{70} v(t)dt$ in terms of the rocket's flight. Use a midpoint Riemann sum with 3 subintervals of equal length to approximate $\int_{10}^{70} v(t)dt$.

Part B

1 point is earned for the explanation

1 point is earned for uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t)dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t)dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

6-2b

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$

17.

t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate

$\int_0^{40} v(t) dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of in $\int_0^{40} v(t) dt$ terms of the plane's flight.

Part A

1 point is earned for $v(5)+v(15)+v(25)+v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10*[v(5)+v(15)+v(25)+v(35)] = 10*[9.2+7.0+2.4+4.3]=229$ The integral gives the total distance in miles that the plane flies during the 40 minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $v(5)+v(15)+v(25)+v(35)$

1 point is earned for the answer

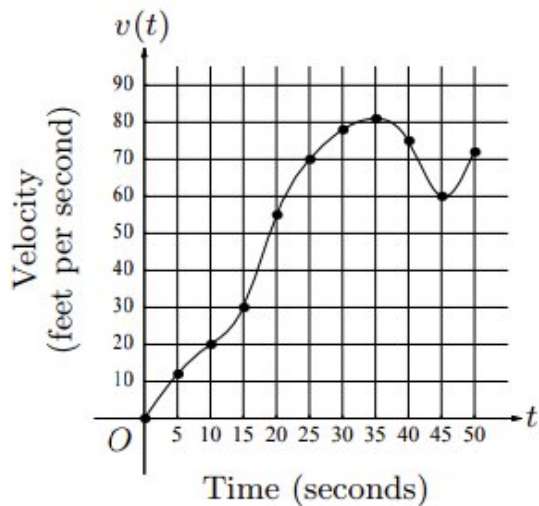
1 point is earned for the meaning with units

Midpoint Riemann sum is $10*[v(5)+v(15)+v(25)+v(35)] = 10*[9.2+7.0+2.4+4.3]=229$

The integral gives the total distance in miles that the plane flies during the 40 minutes.

6-2b

18.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph

Approximate $\int_0^{50} v(t) dt$ with a Riemann sum, using the midpoints of five subintervals of equal length. Using correct units, explain the meaning of this integral.

Part D

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\int_0^{50} v(t) dt$$

$$\approx 10[v(5) + v(15) + v(25) + v(35) + v(45)]$$

$$= 10(12 + 30 + 70 + 81 + 60)$$

6-2b

=2530 feet

This integral is the total distance traveled in feet over the time 0 to 50 seconds.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\int_0^{50} v(t) dt$$

$$\approx 10[v(5) + v(15) + v(25) + v(35) + v(45)]$$

$$= 10(12 + 30 + 70 + 81 + 60)$$

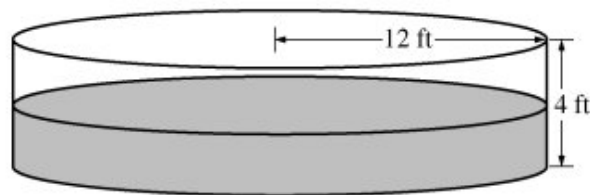
$$= 2530 \text{ feet}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.

6-2b

19.

t	0	2	4	6	8	10	12
$P(t)$	0	46	53	57	60	62	63



The figure above shows an above ground swimming pool in the shape of a cylinder with a radius of 12 feet and a height of 4 feet. The pool contains 1000 cubic feet of water at time $t=0$. During the time interval $0 \leq t \leq 12$ hours, water is pumped into the pool at the rate $P(t)$ cubic feet per hour. The table above gives values of $P(t)$ for selected values of t . During the same time interval, water is leaking from the pool at the rate $R(t)$ cubic feet per hour, where $R(t) = 25e^{-0.05t}$.

(Note: The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

Use a midpoint Riemann sum with three subintervals of equal length to approximate the total amount of water that was pumped into the pool during the time interval $0 \leq t \leq 12$ hours. Show the computations that lead to your answer.

Part A

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} p(t) dt \approx 46.4 + 57.4 + 62.4 = 660 ft^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} p(t) dt \approx 46.4 + 57.4 + 62.4 = 660 ft^3$$

6-2b

20.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, gallons per hour, is given by a differentiable function R of time t . The table above shows the rate as measured every 3 hours for a 24-hour period.

Use a midpoint Riemann sum with 4 subdivisions of equal length to approximate $\int_0^{24} R(t)dt$. Using correct units, explaining the meaning of your answer in terms of water flow.

Part A

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

$$\int_0^{24} R(t)dt \approx 6 [R(3) + R(9) + R(15) + R(21)]$$

$$6 [10.4 + 11.2 + 11.3 + 10.2]$$

$$258.6 \text{ gallons}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

6-2b

$$\int_0^{24} R(t) dt \approx 6 [R(3) + R(9) + R(15) + R(21)]$$
$$6 [10.4 + 11.2 + 11.3 + 10.2]$$
$$258.6 \text{ gallons}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

6-2b

21.  The rate of consumption of cola in the United States is given by $S(t) = Ce^{kt}$, where S is measured in billions of gallons per year and t is measured in years from the beginning of 1980.
- (a) The consumption rate doubles every 5 years and the consumption rate at the beginning of 1980 was 6 billion gallons per year. Find C and k .
- (b) Find the average rate of consumption of cola over the 10-year time period beginning January 1, 1983. Indicate units of measure.
- (c) Use the trapezoidal rule with four equal subdivisions to estimate $\int_5^7 S(t)dt$.
- (d) Using correct units, explain the meaning of $\int_5^7 S(t)dt$ in terms of cola consumption.

Part A

1 point is earned for the correct answer $C = 6$

$$S(t) = Ce^{kt}$$

$$S(0) = 6 \Rightarrow C = 6$$

1 point is earned for the correct answer $12 = 6e^{5k}$

$$S(5) = 12 \Rightarrow 12 = 6e^{5k}$$

1 point is earned for the correct answer $k = \frac{\ln 2}{5}$

$$2 = e^{5k}$$

$$k = \frac{\ln 2}{5}$$

(.138 or .139)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $C = 6$

6-2b

$$S(t) = Ce^{kt}$$

$$S(0) = 6 \Rightarrow C = 6$$

1 point is earned for the correct answer $12 = 6e^{5k}$

$$S(5) = 12 \Rightarrow 12 = 6e^{5k}$$

1 point is earned for the correct answer $k = \frac{\ln 2}{5}$

$$2 = e^{5k}$$

$$k = \frac{\ln 2}{5}$$

(.138 or .139)

Part B

1 point is earned for correctly using $[3,13]$ and dividing by $13 - 3$

1 point is earned for the correct integrand

$$\text{Average rate} = \frac{1}{13 - 3} \int_3^{13} 6e^{\frac{\ln 2}{5}t} dt$$

1 point is earned for the correct answer with units

$$0/1 \text{ if not } \frac{1}{b - a} \int_a^b S(t) dt$$

$$= \frac{3}{\ln 2} [e^{2.6 \ln 2} - e^{0.6 \ln 2}] \text{ billion gal/yr}$$

(19.680 billion gal/yr)

6-2b

Student solves $\frac{S(13) - S(3)}{10} = S'(t)$ for t_0 .

$$S(t_0) = 19.680 \text{ billions gal/yr}$$

3/3 if correct.

Otherwise, read by standard where max. 1-0-0.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $[3, 13]$ and dividing by $13 - 3$

1 point is earned for the correct integrand

$$\text{Average rate} = \frac{1}{13 - 3} \int_3^{13} 6e^{\frac{\ln 2}{5}t} dt$$

1 point is earned for the correct answer with units

$$0/1 \text{ if not } \frac{1}{b - a} \int_a^b S(t) dt$$

$$= \frac{3}{\ln 2} [e^{2.6 \ln 2} - e^{0.6 \ln 2}] \text{ billion gal/yr}$$

$$(19.680 \text{ billion gal/yr})$$

Student solves $\frac{S(13) - S(3)}{10} = S'(t)$ for t_0 .

$$S(t_0) = 19.680 \text{ billions gal/yr}$$

3/3 if correct.

Otherwise, read by standard where max. 1-0-0.

Part C

1 point is earned for the correct trapezoidal rule with S , $n = 4$, interval $[5, 7]$

6-2b

$$\int_5^7 S(t) dt \doteq \frac{1}{4}[S(5) + 2S(5.5) + 2S(6) + 2S(6.5) + S(7)]$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct trapezoidal rule with S , $n = 4$, interval $[5, 7]$

$$\int_5^7 S(t) dt \doteq \frac{1}{4}[S(5) + 2S(5.5) + 2S(6) + 2S(6.5) + S(7)]$$

Part D

1 point is earned for the correct total consumption in a time period

1 point is earned for the correct time period and liquid measure

This gives the total consumption, in billions of gallons, during the years 1985 and 1986.

0/2: "rate of consumption"



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct total consumption in a time period

1 point is earned for the correct time period and liquid measure

This gives the total consumption, in billions of gallons, during the years 1985 and 1986.

0/2: "rate of consumption"

6-2b

22. The rate of consumption of cola in the United States is given by $S(t) = Ce^{kt}$, where S is measured in billions of gallons per year and t is measured in years from the beginning of 1980.

Use the trapezoidal rule with four equal subdivisions to estimate $\int_5^7 S(t) dt$.

Part C

1 point is earned for the correct trapezoidal rule with 5, $n = 4$, interval $[5, 7]$

$$\int_5^7 S(t) dt \doteq \frac{1}{4}[S(5) + 2S(5.5) + 2S(6) + 2S(6.5) + S(7)]$$



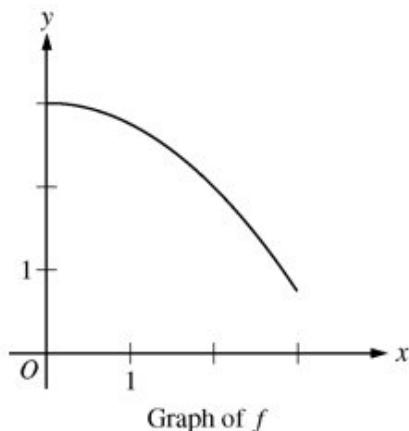
0	1
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The student response earns one of the following points:

1 point is earned for the correct trapezoidal rule with 5, $n = 4$, interval $[5, 7]$

$$\int_5^7 S(t) dt \doteq \frac{1}{4}[S(5) + 2S(5.5) + 2S(6) + 2S(6.5) + S(7)]$$

23.



The graph of the function f is shown above for $0 \leq x \leq 3$. Of the following, which has the least value?

(A) $\int_1^3 f(x) dx$

(B) Left Riemann sum approximation of $\int_1^3 f(x) dx$ with 4 subintervals of equal length

(C) Right Riemann sum approximation of $\int_1^3 f(x) dx$ with 4 subintervals of equal length

(D) Midpoint Riemann sum approximation of $\int_1^3 f(x) dx$ with 4 subintervals of equal length

(E) Trapezoidal sum approximation of $\int_1^3 f(x) dx$ with 4 subintervals of equal length



6-2b

24.



t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

Concert tickets went on sale at noon ($t = 0$) and were sold out within 9 hours. The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Values of $L(t)$ at various times t are shown in the table above.

- (a) Use the data in the table to estimate the rate at which the number of people waiting in line was changing at 5:30 P.M. ($t = 5.5$). Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a trapezoidal sum with three subintervals to estimate the average number of people waiting in line during the first 4 hours that tickets were on sale.
- (c) For $0 \leq t \leq 9$, what is the fewest number of times at which $L'(t)$ must equal 0? Give a reason for your answer.
- (d) The rate at which tickets were sold for $0 \leq t \leq 9$ is modeled by $r(t) = 550te^{-t/2}$ tickets per hour. Based on the model, how many tickets were sold by 3 P.M. ($t = 3$), to the nearest whole number?

Part A

1 point is earned for estimation

1 point is earned for units

$$L'(5.5) = \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour}$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for units

$$L'(5.5) = \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for answer

The average number of people waiting in line during the first 4 hours is approximately

6-2b

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right)$$

= 155.25 people



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for answer

The average number of people waiting in line during the first 4 hours is approximately

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right)$$

= 155.25 people

Part C

1 point is earned for considering change in sign of L'

1 point is earned for the analysis

1 point is earned for the correct conclusion

L is differentiable $[0, 9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1, 3)$ and some t in $(4, 7)$. Similarly, $L'(t) < 0$ for some t in $(3, 4)$ and some t in $(7, 8)$. Then, since L' is continuous on $[0, 9]$, the Intermediate Value Theorem implies that $L'(t) = 0$ for at least three values of t in $[0, 9]$.

OR

1 point is earned for considering relative extrema of L on $(0, 9)$

1 point is earned for the analysis

1 point is earned for the correct conclusion

The continuity of L on $[1, 4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1, 4)$. Similarly, L attains a minimum on $(3, 7)$ and a maximum on $(4, 8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0, 9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0, 9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t) = 0$ for exactly three values of t .]

6-2b



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for considering change in sign of L'

1 point is earned for the analysis

1 point is earned for the correct conclusion

L is differentiable $[0, 9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1, 3)$ and some t in $(4, 7)$. Similarly, $L'(t) < 0$ for some t in $(3, 4)$ and some t in $(7, 8)$. Then, since L' is continuous on $[0, 9]$, the Intermediate Value Theorem implies that $L'(t) = 0$ for at least three values of t in $[0, 9]$.

OR

1 point is earned for considering relative extrema of L on $(0, 9)$

1 point is earned for the analysis

1 point is earned for the correct conclusion

The continuity of L on $[1, 4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1, 4)$. Similarly, L attains a minimum on $(3, 7)$ and a maximum on $(4, 8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0, 9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0, 9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t) = 0$ for exactly three values of t .]

Part D

1 point is earned for the correct integrand

1 point is earned for the correct limits and answer

$$\int_0^3 r(t) dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the correct integrand

1 point is earned for the correct limits and answer

6-2b

$$\int_0^3 r(t) \, dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.

6-2b

25.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43

 As a pot of tea cools, the temperature of the tea is modeled by a differentiable function H for $0 \leq t \leq 10$, where time t is measured in minutes and temperature $H(t)$ is measured in degrees Celsius. Values of $H(t)$ at selected values of time t are shown in the table above.

(a) Use the data in the table to approximate the rate at which the temperature of the tea is changing at time $t = 3.5$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\frac{1}{10} \int_0^{10} H(t) dt$ in the context of this problem. Use a trapezoidal sum with the four subintervals indicated by the table to estimate $\frac{1}{10} \int_0^{10} H(t) dt$.

(c) Evaluate $\int_0^{10} H'(t) dt$. Using correct units, explain the meaning of the expression in the context of this problem.

(d) At time $t = 0$, biscuits with temperature 100°C were removed from an oven. The temperature of the biscuits at time t is modeled by a differentiable function B for which it is known that $B'(t) = -13.84e^{-0.173t}$. Using the given models, at time $t = 10$, how much cooler are the biscuits than the tea?

Part A

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$

Part B

6-2b

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\frac{1}{10} \int_0^{10} H(t) dt \approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ = 52.95$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\frac{1}{10} \int_0^{10} H(t) dt \approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ = 52.95$$

Part C

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

6-2b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

Part D

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer $B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275$; $H(10) - H(0) = 8.817$

The biscuits are 8.817 degrees Celsius cooler than the tea.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer

$$B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275$$
; $H(10) - H(0) = 8.817$

The biscuits are 8.817 degrees Celsius cooler than the tea.

6-2b

26.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t = 0$) and 8 P.M. ($t = 8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

(a) Use the data in the table to approximate the rate, in hundreds of entries per hour, at which entries were being deposited at time $t = 6$. Show the computations that lead to your answer.

(b) Use a trapezoidal sum with the four subintervals given by the table to approximate the value of $18 \int_0^8 E(t) dt$.

. Using correct units, explain the meaning of $18 \int_0^8 E(t) dt$ in terms of the number of entries.

(c) At 8 P.M., volunteers began to process the entries. They processed the entries at a rate modeled by the function P , where $P(t) = t^3 - 30t^2 + 298t - 976$ hundreds of entries per hour for $8 \leq t \leq 12$. According to the model, how many entries had not yet been processed by midnight ($t = 12$)?

(d) According to the model from part (c), at what time were the entries being processed most quickly? Justify your answer.

Part A

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

6-2b

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688\end{aligned}$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688\end{aligned}$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$

Part C

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$



0	1	2
---	---	---

6-2b

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$

Part D

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.

6-2b

27.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

- (a) Estimate $T'(7)$. Show the work that leads to your answer. Indicate units of measure.
- (b) Write an integral expression in terms of $T(x)$ for the average temperature of the wire. Estimate the average temperature of the wire using a trapezoidal sum with the four subintervals indicated by the data in the table. Indicate units of measure.
- (c) Find $\int_0^8 T'(x)dx$, and indicate units of measure. Explain the meaning of $\int_0^8 T'(x)dx$ in terms of the temperature of the wire.
- (d) Are the data in the table consistent with the assertion that $T''(x) > 0$ for every x in the interval $0 < x < 8$? Explain your answer.

Part A

1 point is earned for the answer

$$\frac{T(8) - T(6)}{8 - 6} = \frac{55 - 62}{2} = -\frac{7}{2}^{\circ}\text{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C/cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{T(8) - T(6)}{8 - 6} = \frac{55 - 62}{2} = -\frac{7}{2}^{\circ}\text{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C/cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part B

6-2b

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\int_0^8 T(x) dx$:

$$A = \frac{100 + 93}{2} \cdot 1 + \frac{93 + 70}{2} \cdot 4 + \frac{70 + 62}{2} \cdot 1 + \frac{62 + 55}{2} \cdot 2$$

$$\text{Average temperature} \approx \frac{1}{8} A = 75.6875^\circ \text{C}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^\circ\text{C}/\text{cm}$ in (a), and $^\circ\text{C}$ in (b) and (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\int_0^8 T(x) dx$:

$$A = \frac{100 + 93}{2} \cdot 1 + \frac{93 + 70}{2} \cdot 4 + \frac{70 + 62}{2} \cdot 1 + \frac{62 + 55}{2} \cdot 2$$

6-2b

$$\text{Average temperature} \approx \frac{1}{8}A = 75.6875^{\circ}\text{C}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part C

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire to the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire to the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part D

1 point is earned for the two slopes of secant lines

6-2b

1 point is earned for the answer with explanation

Average rate of change of temperature on $[1, 5]$ is $\frac{70 - 93}{5 - 1} = -5.75$.

Average rate of change of temperature on $[5, 6]$ is $\frac{62 - 70}{6 - 5} = -8$.

No. By the MVT, $T'(c_1) = -5.75$ for some c_1 in the interval $(1, 5)$ and $T'(c_2) = -8$ for some c_2 in the interval $(5, 6)$. It

follows that T' must decrease somewhere in the interval (c_1, c_2) . Therefore T'' is not positive for every x in $[0, 8]$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on $[1, 5]$ is $\frac{70 - 93}{5 - 1} = -5.75$.

Average rate of change of temperature on $[5, 6]$ is $\frac{62 - 70}{6 - 5} = -8$.

No. By the MVT, $T'(c_1) = -5.75$ for some c_1 in the interval $(1, 5)$ and $T'(c_2) = -8$ for some c_2 in the interval $(5, 6)$. It

follows that T' must decrease somewhere in the interval (c_1, c_2) . Therefore T'' is not positive for every x in $[0, 8]$.

6-2b

28.

t (minutes)	0	2	5	7	11	12
$r'(t)$ (feet per minute)	5.7	4.0	2.0	1.2	0.6	0.5

The volume of a spherical hot air balloon expands as the air inside the balloon is heated. The radius of the balloon, in feet, is modeled by a twice-differentiable function r of time t , where t is measured in minutes. For $0 < t < 12$, the graph of r is concave down. The table above gives selected values of the rate of change, $r'(t)$, of the radius of the balloon over the time interval $0 \leq t \leq 12$. The radius of the balloon is 30 feet when $t = 5$.

(Note: The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.)

- (a) Estimate the radius of the balloon when $t = 5.4$ using the tangent line approximation at $t = 5$. Is your estimate greater than or less than the true value? Give a reason for your answer.
- (b) Find the rate of change of the volume of the balloon with respect to time when $t = 5$. Indicate units of measure.
- (c) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_0^{12} r'(t) dt$. Using correct units, explain the meaning of $\int_0^{12} r'(t) dt$ in terms of the radius of the balloon.
- (d) Is your approximation in part (c) greater than or less than $\int_0^{12} r'(t) dt$? Give a reason for your answer.

Part A

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.

Part B

2 points are earned for : $\frac{dV}{dt}$

6-2b

1 point is earned for the answer

$$\frac{dV}{dt} = 3 \left(\frac{4}{3} \right) \pi r^2 \frac{dr}{dt}$$

$$\frac{dV}{dt} \Big|_{t=5} = 4\pi(30)^2 2 = 7200\pi \text{ ft}^3/\text{min}$$

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for : $\frac{dV}{dt}$

1 point is earned for the answer

$$\frac{dV}{dt} = 3 \left(\frac{4}{3} \right) \pi r^2 \frac{dr}{dt}$$

$$\frac{dV}{dt} \Big|_{t=5} = 4\pi(30)^2 2 = 7200\pi \text{ ft}^3/\text{min}$$

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)

Part C

1 point is earned for the approximation

1 point is earned for the explanation

$$\begin{aligned} \int_0^{12} r'(t) dt &\approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5) \\ &= 19.3 \text{ ft} \end{aligned}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

6-2b

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5)$$

$$= 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)

Part D

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$



0	1
---	---

The student response earns all of the following points:

6-2b

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$

6-2b

29.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

(a) Use the data in the table to estimate the value of $v'(16)$.

(b) Using correct units, explain the meaning of the definite integral $\int_0^{40} |v(t)| dt$ in the context of the problem.

Approximate the value of $\int_0^{40} |v(t)| dt$ using a right Riemann sum with the four subintervals indicated in the table.

(c) Bob is riding his bicycle along the same path. For $0 \leq t \leq 10$, Bob's velocity is modeled by $B(t) = t^3 - 6t^2 + 300$, where t is measured in minutes and $B(t)$ is measured in meters per minute. Find Bob's acceleration at time $t = 5$.

(d) Based on the model B from part (c), find Bob's average velocity during the interval $0 \leq t \leq 10$.

Part A

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$

Part B

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

6-2b

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$

Part C

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$

6-2b

Part D

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}\end{aligned}$$

30.

x	2	3	5	8	13
$f(x)$	6	-2	-1	3	9

The function f is continuous on the closed interval $[2, 13]$ and has values as shown in the table above. Using the intervals $[2, 3]$, $[3, 5]$, $[5, 8]$, and $[8, 13]$ what is the approximation of $\int_2^{13} f(x) dx$ obtained from a left Riemann sum?

(A) 6

(B) 14

(C) 28

(D) 32

(E) 50



6-2b

31.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

Use a left Riemann sum with subintervals indicated by the data in the table to approximate $\int_2^{13} f(x) dx$. Show the work that leads to your answer.

Part C

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3-2) + f(3)(5-3) + f(5)(8-5) + f(8)(13-8) = 18$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3-2) + f(3)(5-3) + f(5)(8-5) + f(8)(13-8) = 18$$

6-2b

32. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

Use the trapezoidal rule with four equal subdivisions of the closed interval $[0,1]$ to approximate $F(1)$.

Part A

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or $x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin(\frac{1}{4})^2 + 2 \sin(\frac{1}{2})^2 + 2 \sin(\frac{3}{4})^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or $x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

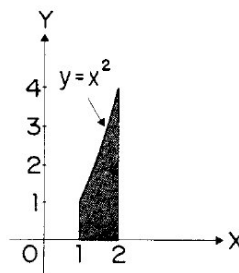
$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin(\frac{1}{4})^2 + 2 \sin(\frac{1}{2})^2 + 2 \sin(\frac{3}{4})^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$

6-2b

33.



Calculate the approximate area of the shaded region in the figure by the trapezoidal rule, using divisions at $x = \frac{4}{3}$ and $x = \frac{5}{3}$.

(A) $\frac{50}{27}$

(B) $\frac{251}{108}$

(C) $\frac{7}{3}$

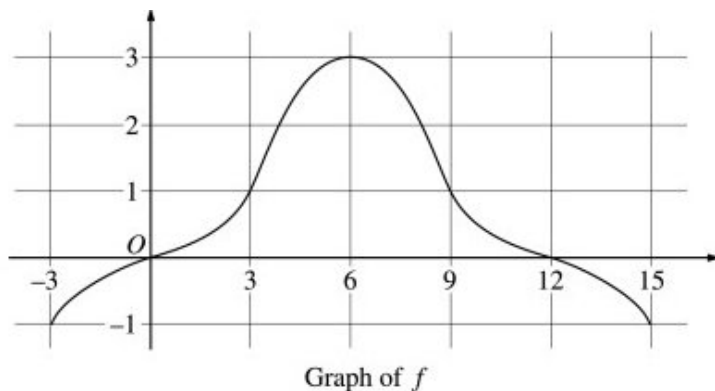
(D) $\frac{127}{54}$

(E) $\frac{77}{27}$



6-2b

34.



The graph of a differentiable function f on the closed interval $[-3, 15]$ is shown in the figure above. The graph of f has a horizontal tangent line at $x = 6$. Let $g(x) = 5 + \int_6^x f(t)dt$ for $-3 \leq x \leq 15$.

Find a trapezoidal approximation of $\int_{-3}^{15} f(t)dt$ using six subintervals of length $\Delta t = 3$.

Part D

1 point is earned for the trapezoidal method

$$\frac{3}{2}(-1 + 2(0 + 1 + 3 + 1 + 0) - 1) = 12$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the trapezoidal method

$$\frac{3}{2}(-1 + 2(0 + 1 + 3 + 1 + 0) - 1) = 12$$

6-2b

35.

Distance from the river's edge (feet)	0	8	14	22	24
Depth of the water (feet)	0	7	8	2	0

A scientist measures the depth of the Doe River at Picnic Point. The river is 24 feet wide at this location. The measurements are taken in a straight line perpendicular to the edge of the river. The data are shown in the table above. The velocity of the water at Picnic Point, in feet per minute, is modeled by $v(t) = 16 + 2 \sin(\sqrt{t+10})$ for $0 \leq t \leq 120$ minutes.

Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate the area of the cross section of the river at Picnic Point, in square feet. Show the computations that lead to your answer.

Part A

1 point is earned for trapezoidal approximation

$$\frac{(0+7)}{2} \cdot 8 + \frac{(7+8)}{2} \cdot 6 + \frac{(8+2)}{2} \cdot 8 + \frac{(2+0)}{2} \cdot 2$$

$$=115 \text{ ft}^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for trapezoidal approximation

$$\frac{(0+7)}{2} \cdot 8 + \frac{(7+8)}{2} \cdot 6 + \frac{(8+2)}{2} \cdot 8 + \frac{(2+0)}{2} \cdot 2$$

$$=115 \text{ ft}^2$$

36.

x	0	0.5	1.0	1.5	2.0
$f(x)$	3	3	5	8	13

A table of values for a continuous function f is shown above. If four equal subintervals of $[0,2]$ are used, which of the following is the trapezoidal approximation of $\int_0^2 f(x) dx$?

(A) 8

(B) 12

(C) 16

(D) 24

(E) 32



6-2b

37.

x	2	5	10	14
$f(x)$	12	28	34	30

The function f is continuous on the closed interval $[2,14]$ and has values as shown in the table above. Using the subintervals $[2,5]$, $[5,10]$, and $[10,14]$, what is the approximation of $\int_2^{14} f(x)dx$ found by using a right Riemann sum?

(A) 296

(B) 312

(C) 343

(D) 374

(E) 390



38.

x	2	5	7	8
$f(x)$	10	30	40	20

The function f is continuous on the closed interval $[2,8]$ and has values that are given in the table above. Using the subintervals $[2,5]$, $[5,7]$, and $[7,8]$, what is the trapezoidal approximation of $\int_2^8 f(x)dx$?

(A) 110

(B) 130

(C) 160

(D) 190

(E) 210



39.

t (sec)	0	2	4	6
$a(t)$ (ft/sec ²)	5	2	8	3

The data for the acceleration $a(t)$ of a car from 0 to 6 seconds are given in the table above. If the velocity at $t=0$ is 11 feet per second, the approximate value of the velocity at $t=6$, computed using a left-hand Riemann sum with three subintervals of equal length, is

(A) 26 ft/sec

(B) 30 ft/sec

(C) 37 ft/sec

(D) 39 ft/sec

(E) 41 ft/sec



6-2b

40.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t=0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t=0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

Part C

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\frac{1}{20} \int_0^{20} W(t) dt \approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15))$$

$$= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9)$$

$$= \frac{1}{20} \cdot 1215.8 = 60.79$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\frac{1}{20} \int_0^{20} W(t) dt \approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15))$$

6-2b

$$= \frac{1}{20}(4.55.0 + 5.57.1 + 6.61.8 + 5.67.9)$$

$$= \frac{1}{20}.1215.8 = 60.79$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

6-2b

41. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	7	10	12
$r'(t)$ (centimeters per day)	-6.1	-5.0	-4.4	-3.8	-3.5

An ice sculpture melts in such a way that it can be modeled as a cone that maintains a conical shape as it decreases in size. The radius of the base of the cone is given by a twice-differentiable function r , where $r(t)$ is measured in centimeters and t is measured in days. The table shown gives selected values of $r'(t)$, the rate of change of the radius, over the time interval $0 \leq t \leq 12$.

(a) Approximate $r''(8.5)$ using the average rate of change of r' over the interval $7 \leq t \leq 10$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t , $0 \leq t \leq 3$, for which $r'(t) = -6$? Justify your answer.

(c) Use a right Riemann sum with the four subintervals indicated in the table to approximate the value of $\int_0^{12} r'(t) \, dt$.

(d) The height of the cone decreases at a rate of 2 centimeters per day. At time $t = 3$ days, the radius is 100 centimeters and the height is 50 centimeters. Find the rate of change of the volume of the cone with respect to time, in cubic centimeters per day, at time $t = 3$ days. (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the supporting work must include at least a difference and a quotient.

Simplification is not required to earn the first point. If the numerical value is simplified, it must be correct.

The second point can be earned with an incorrect approximation for $r''(8.5)$ but cannot be earned without some value for $r''(8.5)$ presented.

Units may be written in any equivalent form (such as cm/day^2).

6-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $r''(8.5)$ with supporting work
- ☐ Units

Solution:

$$r''(8.5) \approx \frac{r'(10) - r'(7)}{10 - 7} = \frac{-3.8 - (-4.4)}{10 - 7} \text{ centimeter per day per day}$$

$$= \frac{0.6}{3} = 0.2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, the response must establish that -6 is between $r'(0)$ and $r'(3)$ (or -6.1 and -5). This statement may be represented symbolically (with or without including one or both endpoints in an inequality) or verbally. A response of “ $r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$ ” does not state that -6 is between -6.1 and -5 . Thus this response does not earn the first point.

To earn the second point:

- The response must state that $r'(t)$ is continuous because $r'(t)$ is differentiable (or because $r(t)$ is twice differentiable).
- The response must have earned the first point. Exception: A response of “ $r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$ ” does not earn the first point because of imprecise communication but may nonetheless earn the second point if all other criteria for the second point are met.
- The response must conclude that there is a time t such that $r'(t) = -6$ (A statement of “yes” would be sufficient.)

To earn the second point, the response need not explicitly name the Intermediate Value Theorem, but if a theorem is named, it must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $r'(0) < -6 < r'(3)$
- ☐ Conclusion using Intermediate Value Theorem

Solution:

6-2b

$r(t)$ is twice-differentiable. $\Rightarrow r'(t)$ is differentiable.

$\Rightarrow r'(t)$ is continuous.

$$r'(0) = -6.1 < -6 < -5.0 = r'(3)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $0 < t < 3$, such that $r'(t) = -6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, at least seven of the eight factors in the Riemann sum must be correct. If there is any error in the Riemann sum, the response does not earn the second point.

A response of $3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5)$ earns both the first and second points, unless there is a subsequent error in simplification, in which case the response would earn only the first point.

A response that presents the correct answer, with accompanying work that shows the four products in the Riemann sum (without explicitly showing all of the factors and/or the sum process) does not earn the first point but earns the second point. For example, $-15 + 4(-4.4) + 3(-3.8) + -7$ does not earn the first point but earns the second point. Similarly, $-15, -17.6, -11.4, -7 \rightarrow -51$ does not earn the first point but earns the second point.

A response that presents the correct answer (-51) with no supporting work earns no points.

A response that provides a completely correct left Riemann sum and approximation $\int_0^{12} r'(t) dt$ (i.e., $3r'(0) + 4r'(3) + 3r'(7) + 2r'(10) = 3(-6.1) + 4(-5.0) + 3(-4.4) + 2(-3.8) = -59.1$) earns 1 of the 2 points.

A response that has any error in a left Riemann sum or evaluation for $\int_0^{12} r'(t) dt$ earns no points.

Units are not required or read in this part.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned}
 \int_0^{12} r'(t) dt &\approx 3r'(3) + 4r'(7) + 3r'(10) + 2r'(12) \\
 &= 3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5) \\
 &= -51
 \end{aligned}$$

Part D

6-2b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first 2 points could be earned in either order.

A response with a completely correct product rule, missing one or both of the correct differentials, earns the product rule point, but not the chain rule point.

For example, $\frac{dV}{dt} = \frac{2}{3}\pi r h + \frac{1}{3}\pi r^2$ earns the first point, but not the second.

A response that treats r or h (but not both) as a constant is eligible for the chain rule point but not the product rule point. For example, $\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt}$ is correct if h is constant, and thus earns the chain rule point.

Note: Neither $\frac{dV}{dt} = \frac{2}{3}\pi r \frac{dh}{dt}$ nor $\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt} \frac{dh}{dt}$ earns any points.

A response that assumes a functional relationship between r and h (such as $r = 2h$), and uses this relationship to create a function for volume in terms of one variable, is eligible for at most the chain rule point. For example,

$$r = 2h \rightarrow V = \frac{1}{3}\pi(2h)^2 h = \frac{4}{3}\pi h^3 \rightarrow \frac{dV}{dt} = 4\pi h^2 \frac{dh}{dt} \text{ earns only the chain rule point.}$$

A response that mishandles the constant $\frac{1}{3}\pi$ cannot earn the third point but is eligible for the first 2 points.

The third point cannot be earned without both of the first 2 points.

$$\frac{dV}{dt} = \frac{2}{3}\pi(100)(50)(-5) + \frac{1}{3}\pi(100)^2(-2) \text{ earns all 3 points.}$$

Units are not required or read in this part.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt} + \frac{1}{3}\pi r^2 \frac{dh}{dt}$$

$$\frac{dV}{dt} \Big|_{t=3} = \frac{2}{3}\pi(100)(50)(-5) + \frac{1}{3}\pi(100)^2(-2) = -\frac{70,000\pi}{3}$$

The rate of change of the volume of the sculpture at $t = 3$ is approximately $-\frac{70,000\pi}{3}$ cubic centimeters per day.

6-2b

42. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	2	4	6	8	10	12
$R(t)$ (vehicles per hour)	2935	3653	3442	3010	3604	1986	2201

On a certain weekday, the rate at which vehicles cross a bridge is modeled by the differentiable function R for $0 \leq t \leq 12$, where $R(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). Values of $R(t)$ for selected values of t are given in the table above.

Use the data in the table to approximate $R'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $R'(5)$ in the context of the problem.

Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\int_0^{12} R(t) \, dt$. Indicate units of measure.

On a certain weekend day, the rate at which vehicles cross the bridge is modeled by the function H defined by $H(t) = -t^3 - 3t^2 + 288t + 1300$ for $0 \leq t \leq 17$, where $H(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). According to this model, what is the average number of vehicles crossing the bridge per hour on the weekend day for $0 \leq t \leq 12$?

For $12 < t < 17$, $L(t)$, the local linear approximation to the function H given in part (c) at $t = 12$, is a better model for the rate at which vehicles cross the bridge on the weekend day. Use $L(t)$ to find the time t , for $12 < t < 17$, at which the rate of vehicles crossing the bridge is 2000 vehicles per hour. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.

The second point requires a time reference, a rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

6-2b



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ approximation using values from table
- ☐ interpretation with units

Solution:

$$R'(5) \approx \frac{R(6) - R(4)}{6 - 4} = \frac{3010 - 3442}{2} = -216$$

At time $t = 5$ hours (12 p.m.), the rate at which vehicles cross the bridge is decreasing at a rate of approximately 216 vehicles per hour per hour.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

- The first point does not require a simplified answer. Substitution of function values is required. A single substitution error for one and only one of $R(2)$, $R(6)$, or $R(10)$ will still earn this point.

The second point requires earning the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ midpoint sum setup
- ☐ approximation using values from table with units

Solution:

$$\begin{aligned} \int_0^{12} R(t) \, dt &\approx 4(R(2) + R(6) + R(10)) \\ &= 4(3653 + 3010 + 1986) \\ &= 34,596 \text{ vehicles} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct integrand and limits of integration. The constant $\frac{1}{12}$ is not needed to earn this point.

6-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\frac{1}{12-0} \int_0^{12} H(t) dt = 2452$$

On the weekend day for $0 \leq t \leq 12$, the average number of vehicles crossing the bridge is 2452 vehicles per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ slope
- ☐ $L(t) = 2000$
- ☐ answer

Solution:

$$L(t) = H(12) + H'(12)(t - 12)$$

$$H(12) = 2596$$

$$H'(12) = -216$$

$$L(t) = 2000 \Rightarrow t = 14.759259$$

On the weekend day, the rate at which vehicles cross the bridge is 2000 vehicles per hour at time $t = 14.759$ hours.

6-2b

43.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

r (centimeters)	0	1	2	2.5	4
$f(r)$ (milligrams per square centimeter)	1	2	6	10	18

The density of a bacteria population in a circular petri dish at a distance r centimeters from the center of the dish is given by an increasing, differentiable function f , where $f(r)$ is measured in milligrams per square centimeter. Values of $f(r)$ for selected values of r are given in the table above.

(a) Use the data in the table to estimate $f'(2.25)$. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) The total mass, in milligrams, of bacteria in the petri dish is given by the integral expression $2\pi \int_0^4 r f(r) \, dr$.

Approximate the value of $2\pi \int_0^4 r f(r) \, dr$ using a right Riemann sum with the four subintervals indicated by the data in the table.

(c) Is the approximation found in part (b) an overestimate or underestimate of the total mass of bacteria in the petri dish? Explain your reasoning.

(d) The density of bacteria in the petri dish, for $1 \leq r \leq 4$, is modeled by the function g defined by $g(r) = 2 - 16(\cos(1.57\sqrt{r}))^3$. For what value of k , $1 < k < 4$, is $g(k)$ equal to the average value of $g(r)$ on the interval $1 \leq r \leq 4$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must provide both a difference and a quotient and must explicitly use values of f from the table.

6-2b

Simplification of the numerical value is not required to earn the first point. If the numerical value is simplified, it must be correct.

The interpretation requires all of the following: distance $r = 2.25$, density of bacteria (population) is increasing or changing, at a rate of 8, and units of milligrams per square centimeter per centimeter.

The second point (interpretation) cannot be earned without a nonzero presented value for $f'(2.25)$.

To earn the second point the interpretation must be consistent with the presented nonzero value for $f'(2.25)$. In particular, if the presented value for $f'(2.25)$ is negative, the interpretation must include “decreasing at a rate of $|f'(2.25)|$ ” or “changing at a rate of $f'(2.25)$.” The second point cannot be earned for an incorrect statement such as “the bacteria density is decreasing at a rate of $-8\dots$ ” even for a presented $f'(2.25) = -8$.

The units (mg/cm²/cm) may be attached to the estimate of $f'(2.25)$ and, if so, do not need to be repeated in the interpretation.

If units attached to the estimate do not agree with units in the interpretation, read the units in the interpretation.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Estimate
- ☐ Interpretation with units

Solution:

$$f'(2.25) \approx \frac{f(2.5) - f(2)}{2.5 - 2} = \frac{10 - 6}{0.5} = 8$$

At a distance of $r = 2.25$ centimeters from the center of the petri dish, the density of the bacteria population is increasing at a rate of 8 milligrams per square centimeter per centimeter.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The presence or absence of 2π has no bearing on earning the first point.

The first point is earned for a sum of four products with at most one error in any single value among the four products. Multiplication by 1 in any term does not need to be shown, but all other products must be explicitly shown.

A response of $1 \cdot f(1) \cdot (1 - 0) + 2 \cdot f(2) \cdot (2 - 1) + 2.5 \cdot f(2.5) \cdot (2.5 - 2) + 4 \cdot f(4) \cdot (4 - 2.5)$ earns the first point but not the second.

A response with any error in the Riemann sum is not eligible for the second point.

6-2b

A response that provides a completely correct left Riemann sum for $2\pi \int_0^4 r f(r) dr$ and approximation (91π) earns one of the two points. A response that has any error in a left Riemann sum or evaluation for $2\pi \int_0^4 r f(r) dr$ earns no points.

A response that provides a completely correct right Riemann sum for $2\pi \int_0^4 f(r) dr$ and approximation (80π) earns one of the two points. A response that has any error in a right Riemann sum or evaluation for $2\pi \int_0^4 f(r) dr$ earns no points.

Simplification of the numerical value is not required to earn the second point. If a numerical value is given, it must be correct to three decimal places.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation

Solution:

$$\begin{aligned}
 2\pi \int_0^4 r f(r) dr &\approx 2\pi(1 \cdot f(1) \cdot (1 - 0) + 2 \cdot f(2) \cdot (2 - 1) \\
 &\quad + 2.5 \cdot f(2.5) \cdot (2.5 - 2) + 4 \cdot f(4) \cdot (4 - 2.5)) \\
 &= 2\pi(1 \cdot 2 \cdot 1 + 2 \cdot 6 \cdot 1 + 2.5 \cdot 10 \cdot 0.5 + 4 \cdot 18 \cdot 1.5) \\
 &= 269\pi = 845.088
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point a response must explain that $r f(r)$ is increasing and, therefore, the right Riemann sum is an overestimate. The second point can be earned without having earned the first point.

A response that attempts to explain based on a left Riemann sum for $2\pi \int_0^4 r f(r) dr$ from part (b) earns no points.

A response that attempts to explain based on a right Riemann sum for $2\pi \int_0^4 f(r) dr$ from part (b) earns no points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

6-2b

- ☐ Product rule expression for $\frac{d}{dr}(rf(r))$
- ☐ Answer with explanation

Solution:

$$\frac{d}{dr}(rf(r)) = f(r) + rf'(r)$$

Because f is nonnegative and increasing, $\frac{d}{dr}(rf(r)) > 0$ on the interval $0 \leq r \leq 4$. Thus, the integrand $rf(r)$ is strictly increasing.

Therefore, the right Riemann sum approximation of $2\pi \int_0^4 rf(r) dr$ is an overestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral, with or without $\frac{1}{4-1}$ or $\frac{1}{3}$.

A response that presents a definite integral with incorrect limits but a correct integrand earns the first point.

Presentation of the numerical value 9.875795 is not required to earn the second point. This point can be earned by the average value setup: $\frac{1}{3} \int_1^4 g(r) dr$.

Once earned for the average value setup, the second point cannot be lost. Subsequent errors will result in not earning the third point.

The third point is earned only for the value $k = 2.497$.

The third point cannot be earned without the second.

Special case: A response that does not provide the average value setup but presents an average value of -13.955 is using degree mode on their calculator. This response would not earn the second point but could earn the third point for an answer of $k = 2.5$ (or 2.499).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Definite integral
- ☐ Average value
- ☐ Answer

Solution:

6-2b

$$\text{Average value} = g_{\text{avg}} = \frac{1}{4-1} \int_1^4 g(r) \, dr$$

$$\frac{1}{4-1} \int_1^4 g(r) \, dr = 9.875795$$

$$g(k) = g_{\text{avg}} \Rightarrow k = 2.497$$

6-2b

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

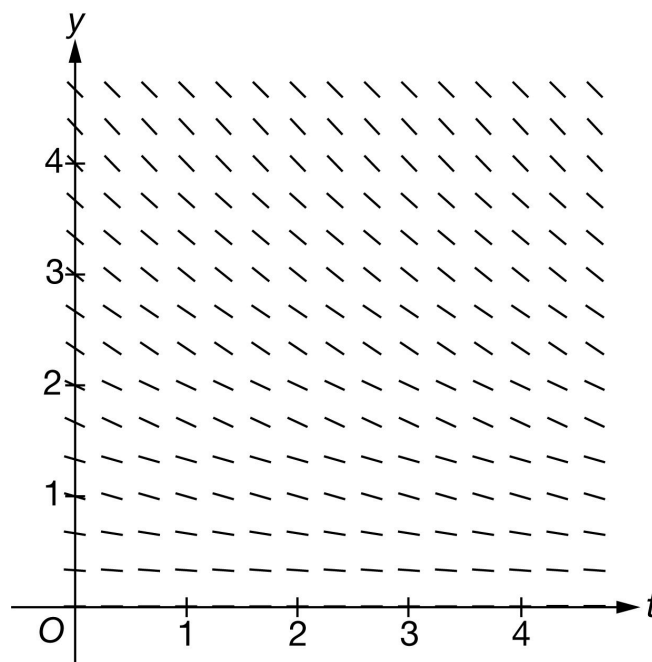
t (days)	0	5	10	15	20
$M(t)$ (number of 1-cent coins)	0	300	500	870	1100

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 20$, the total number of coins in the container is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

- (a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{20} M(t) \, dt$. Show the computations that lead to your answer.
- (b) Find the value of $\int_0^5 M'(3t) \, dt$. Show the computations that lead to your answer.
- (c) Using correct units, approximate $M'(15)$ using the average rate of change of M over the interval $10 \leq t \leq 20$.
- (d) Explain the meaning of $M'(15)$ in the context of the problem.
- (e) It is known that $M'(10) = 55$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 10$.
- (f) It is known that $M''(t) > 0$ for $10 \leq t \leq 15$. If $M(14)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $M(14)$? Give a reason for your answer.
- (g) For $0 \leq t \leq 20$, the total number of coins collected can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each coin is 3.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 11$ days.

6-2b

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 10 \cdot (M(5) + M(15))$$

$$\cdot 10 \cdot M(5) + 10 \cdot M(15)$$

$$\cdot (10 - 0) \cdot 300 + (20 - 10) \cdot 870$$

$$\cdot 10 \cdot (300 + 870) \text{ or } 10 \cdot 300 + 10 \cdot 870$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{10}{b} \cdot (M(5) + M(15))$, for any nonzero value of b would earn the point. Note: The value $\frac{10}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (M(5) + M(15))$ for $a \neq 10$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (Form of midpoint sum) point was earned.

6-2b

Minimal complete answers that earn both points:

$$\cdot 10 \cdot (300 + 870)$$

$$\cdot 10 \cdot 300 + 10 \cdot 870$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{20} M(t) dt \approx M(5)(10 - 0) + M(15)(20 - 10)$$

$$= 300 \cdot 10 + 870 \cdot 10 = 11,700$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int M'(3t) dt$ or $\int M'(u) du$.

The point is earned for any of the following:

· The equation $\int M'(3t) dt = M(3t)$ or $\int M'(u) du = M(u)$

· $\frac{1}{3}M(3t)\big|_0^5$ or $\frac{1}{3}M(u)\big|_0^{15}$

· $M(15) - M(0)$, $M(15)$, or $M(0)$

6-2b

This point can be earned if the substitution point is not.

Note: $\int_a^b M'(u) \, du = M(b) - M(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of M' is M .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$M(15)$ and $M(0)$ must be evaluated; the expression $\frac{1}{3}(M(15) - M(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(870 - 0)$ or $\frac{1}{3} \cdot 870$

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{3}M(3t)\big|_0^5$

· $\frac{1}{3}(M(15) - M(0))$

Responses of only the following would earn only the second point:

· $M(15) - M(0)$

· $M(5) - M(0)$

· $M(15)$

· $\int_0^5 M'(u) \, du = M(5)$

· $M(3t)\big|_0^5$ or $M(u)\big|_0^{15}$

The following responses would earn no points:

· 290 in the absence of one of the first two points.

· $\int_0^5 M'(3t) \, dt = M(5)$



0	1	2	3
---	---	---	---

6-2b

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 15$.

$$\begin{aligned}\int_0^5 M'(3t) \, dt &= \frac{1}{3} \int_0^{15} M'(u) \, du \\ &= \frac{1}{3}(M(15) - M(0)) \\ &= \frac{1}{3}(870 - 0) = 290\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{1100-500}{10}$ or $\frac{600}{20-10}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

6-2b

Solution:

$$M'(15) \approx \frac{M(20) - M(10)}{20 - 10} = \frac{1100 - 500}{10} = 60 \text{ coins per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins in the container
- the specific point in time, $t = 15$

Responses do not need to include correct units or any units.

The word “increasing” may be used in place of the word “changing”.

The response is not earned for “the rate of coins deposited on day 15 “ as this does not indicate a change in the rate of deposit.

“the rate of change of M “ without specifying that M is the number of coins deposited, does not earn the point.

Acceptable ways to provide the specific point in time:

- On day 15 or after 15 days.
- At $t = 15$ (missing the unit of time)
- On week 15 (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 15 days” or “from day 15 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(15)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 15.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-2b

The use of $M(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

$$y = 500 + 55(t - 10)$$

$$y - 500 = 55(t - 10)$$

$$y = 55t - 550 + 500$$

$$y = 55t - 50$$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(10)(t - 10) + M(10)$$

$$y = 55(t - 10) + 500$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $M''(t) > 0$ and underestimate.

“ $M(t)$ is concave up” can be used instead of $M''(t) > 0$.

Information about $M'(t)$ by itself without reference to $M''(t) > 0$ is not sufficient and does not earn this point.

“Underestimate because it is concave up” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $M''(t) > 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $M'(t)$ is concave up, underestimate” does not earn this point.



0	1
---	---

6-2b

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $M''(t) > 0$, the tangent line approximation would produce an underestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $3.5R'(11)$ earns this point.

Only expression $3.5M'(11)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement, for example $R'(11) = 3.5M'(11)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per day, of the total weight of the coins at time $t = 11$ days is $3.5R'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.

6-2b

· “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

- “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.
- “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.
- Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

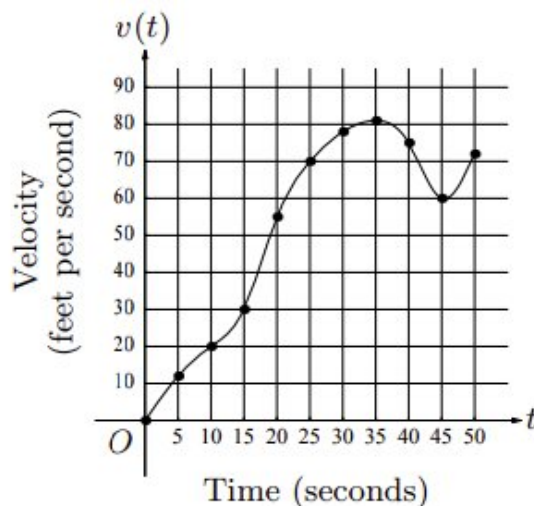
☐ Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-2b

45.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph.

- (a) During what intervals of time is the acceleration of the car positive? Give a reason for your answer
- (b) Find the average acceleration of the car, in ft/sec^2 , over the interval $0 \leq t \leq 50$.
- (c) Find one approximation for the acceleration of the car, in ft/sec^2 , at $t = 40$. Show the computations you used to arrive at your answer.
- (d) Approximate $\int_0^{50} v(t) dt$ with a Riemann sum, using the midpoints of five subintervals of equal length. Using correct units, explain the meaning of this integral.

Part A

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$



0	1	2	3
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The student response earns three of the following points:

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$

6-2b

Part B

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2

Part C

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

6-2b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

Part D

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} & \int_0^{50} v(t) dt \\ & \approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ & = 10(12 + 30 + 70 + 81 + 60) \\ & = 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for midpoint Riemann sum

6-2b

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned}\int_0^{50} v(t) dt \\ &\approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ &= 10(12 + 30 + 70 + 81 + 60) \\ &= 2530 \text{ feet}\end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.

6-2b

46.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	0.3	1.7	2.8	4
$v_P(t)$ (meters per hour)	0	55	-29	55	48

The velocity of a particle, P , moving along the x -axis is given by the differentiable function v_P , where $v_P(t)$ is measured in meters per hour and t is measured in hours. Selected values of $v_P(t)$ are shown in the table above. Particle P is at the origin at time $t = 0$.

- Justify why there must be at least one time t , for $0.3 \leq t \leq 2.8$, at which $v_P'(t)$, the acceleration of particle P , equals 0 meters per hour per hour.
- Use a trapezoidal sum with the three subintervals $[0, 0.3]$, $[0.3, 1.7]$, and $[1.7, 2.8]$ to approximate the value of $\int_0^{2.8} v_P(t) \, dt$.
- A second particle, Q , also moves along the x -axis so that its velocity for $0 \leq t \leq 4$ is given by $v_Q(t) = 45\sqrt{t} \cos(0.063t^2)$ meters per hour. Find the time interval during which the velocity of particle Q is at least 60 meters per hour. Find the distance traveled by particle Q during the interval when the velocity of particle Q is at least 60 meters per hour.
- At time $t = 0$, particle Q is at position $x = -90$. Using the result from part (b) and the function v_Q from part (c), approximate the distance between particles P and Q at time $t = 2.8$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for expressing $v_P(2.8) - v_P(0.3) = 0$. Because these values come from the table, expressing $55 - 55$ earns the first point.

The second point is earned for declaring v_P to be continuous and connecting $v_P'(t)$ to the work for the first point.

6-2b

Another approach involves an Extreme Value Theorem argument. The first point is earned for declaring $v_P(0.3) > v_P(1.7)$ and $v_P(1.7) < v_P(2.8)$. The second point is earned for declaring v_P to be continuous, and using continuity to conclude that $v_P(t)$ must have a minimum at some point c , $0.3 < c < 2.8$, so $v_P'(c) = 0$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $v_P(2.8) - v_P(0.3) = 0$
- ☐ justification, using Mean Value Theorem

v_P is differentiable $\Rightarrow v_P$ is continuous on $0.3 \leq t \leq 2.8$.

$$\frac{v_P(2.8) - v_P(0.3)}{2.8 - 0.3} = \frac{55 - 55}{2.5} = 0$$

By the Mean Value Theorem, there is a value c , $0.3 < c < 2.8$, such that $v_P'(c) = 0$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Simplification is not required to earn this point. The point requires substitution of function values from the table.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ answer, using trapezoidal sum

Solution:

$$\begin{aligned} \int_0^{2.8} v_P(t) dt &\approx 0.3 \left(\frac{v_P(0) + v_P(0.3)}{2} \right) + 1.4 \left(\frac{v_P(0.3) + v_P(1.7)}{2} \right) \\ &\quad + 1.1 \left(\frac{v_P(1.7) + v_P(2.8)}{2} \right) \\ &= 0.3 \left(\frac{0 + 55}{2} \right) + 1.4 \left(\frac{55 + (-29)}{2} \right) + 1.1 \left(\frac{-29 + 55}{2} \right) \\ &= 40.75 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-2b

The first point is earned for an open or closed interval. The interval must be stated explicitly analytically or verbally; using the endpoints of the interval as the limits of integration in a definite integral is not sufficient to earn the first point.

The second point is earned for a definite integral of the form $\int_{t_0}^{t_1} v_Q(t) dt$ or $\int_{t_0}^{t_1} |v_Q(t)| dt$.

The third point does not require units.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ interval
- ☐ definite integral
- ☐ distance

Solution:

$$v_Q(t) = 60 \Rightarrow t = A = 1.866181 \text{ or } t = B = 3.519174$$

$$v_Q(t) \geq 60 \text{ for } A \leq t \leq B$$

$$\int_A^B |v_Q(t)| dt = 106.108754$$

The distance traveled by particle Q during the interval $A \leq t \leq B$ is 106.109 (or 106.108) meters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the integral $\int_0^{2.8} v_Q(t) dt$ or $\int_0^{2.8} |v_Q(t)| dt$.

The third point is earned for the difference between $x_P(2.8)$ and $x_Q(2.8)$. Any incorrect value from Part B of this question, handled correctly, makes the response eligible for the third point. The third point does not require units.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\int_0^{2.8} v_Q(t) dt$
- ☐ position of particle Q

6-2b☐ answer**Solution:**

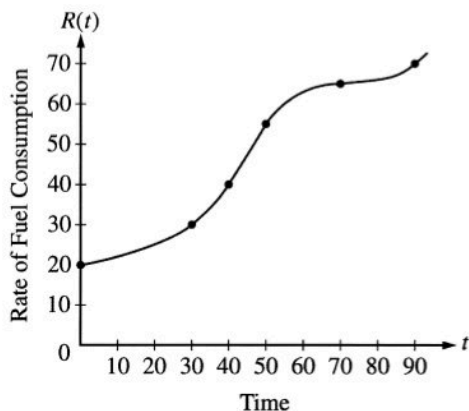
From Part B, the position of particle P at time $t = 2.8$ is $x_P(2.8) = \int_0^{2.8} v_P(t) dt \approx 40.75$.

$$x_Q(2.8) = x_Q(0) + \int_0^{2.8} v_Q(t) dt = -90 + 135.937653 = 45.937653$$

Therefore at time $t = 2.8$, particles P and Q are approximately $45.937653 - 40.75 = 5.188$ (or 5.187) meters apart.

6-2b

47.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

(a) Use data from the table to find an approximation for $R'(45)$. Show the computations that lead to your answer. Indicate units of measure.

(b) The rate of fuel consumption is increasing fastest at time $t = 45$ minutes. What is the value of $R''(45)$? Explain your reasoning.

(c) Approximate the value of $\int_0^{90} R(t) dt$ using a left Riemann sum with the five subintervals indicated by the data in the table. Is this numerical approximation less than the value of $\int_0^{90} R(t) dt$? Explain your reasoning.

(d) For $0 < b \leq 90$ minutes, explain the meaning of $\int_0^b R(t) dt$ in terms of fuel consumption for the plane. Explain the meaning of $\frac{1}{b} \int_0^b R(t) dt$ in terms of fuel consumption for the plane. Indicate units of measure in both answers.

Part A

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

6-2b

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units

Part B

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.

Part C

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

6-2b

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.

Part D

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers

6-2b

48.

 Distance from the river's edge (feet)	0	8	14	22	24
Depth of the water (feet)	0	7	8	2	0

A scientist measures the depth of the Doe River at Picnic Point. The river is 24 feet wide at this location. *The measurements are taken in a straight line perpendicular to the edge of the river. The data are shown in the table above. The velocity of the water at Picnic Point, in feet per minute, is modeled by $v(t) = 16 + 2 \sin(\sqrt{t+10})$ for $0 \leq t \leq 120$ minutes.*

(a) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate the area of the cross section of the river at Picnic Point, in square feet. Show the computations that lead to your answer.

(b) The volumetric flow at a location along the river is the product of the cross-sectional area and the velocity of the water at that location. Use your approximation from part (a) to estimate the average value of the volumetric flow at Picnic Point, in cubic feet per minute, from $t = 0$ to $t = 120$ minutes.

(c) The scientist proposes the function f , given by $f(x) = 8\sin(\pi x/24)$, as a model for the depth of the water, in feet, at Picnic Point x feet from the river's edge. Find the area of the cross section of the river at Picnic Point based on this model.

(d) Recall that the volumetric flow is the product of the cross-sectional area and the velocity of the water at a location. To prevent flooding, water must be diverted if the average value of the volumetric flow at Picnic Point exceeds 2100 cubic feet per minute for a 20-minute period. Using your answer from part (c), find the average value of the volumetric flow during the time interval $40 \leq t \leq 60$ minutes. Does this value indicate that the water must be diverted?

Part A

1 point is earned for trapezoidal approximation

$$\frac{(0+7)}{2} \cdot 8 + \frac{(7+8)}{2} \cdot 6 + \frac{(8+2)}{2} \cdot 8 + \frac{(2+0)}{2} \cdot 2$$

$$= 115 \text{ ft}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for trapezoidal approximation

$$\frac{(0+7)}{2} \cdot 8 + \frac{(7+8)}{2} \cdot 6 + \frac{(8+2)}{2} \cdot 8 + \frac{(2+0)}{2} \cdot 2$$

$$= 115 \text{ ft}^2$$

Part B

1 point is earned for the limits and average value of constant

6-2b

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} 115v(t) dt$$

$$= 1807.169 \text{ or } 1807.170 \text{ ft}^3/\text{min}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} 115v(t) dt$$

$$= 1807.169 \text{ or } 1807.170 \text{ ft}^3/\text{min}$$

Part C

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

6-2b

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$

Part D

1 point is earned for the volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}.$$

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds $2100 \text{ ft}^3/\text{min}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}.$$

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds $2100 \text{ ft}^3/\text{min}$.

6-2b

49.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

(a) Find the average acceleration of rocket A over the time interval $0 \leq t \leq 80$ seconds. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_{10}^{70} v(t) dt$ in terms of the rockets flight. Use a midpoint Riemann sum with 3 subintervals of equal length to approximate $\int_{10}^{70} v(t) dt$.

(c) Rocket B is launched upward with an acceleration of $a(t) = \frac{3}{\sqrt{t+1}}$ feet per second per second. At time $t = 0$ seconds, the initial height of the rocket is 0 feet, and the initial velocity is 2 feet per second. Which of the two rockets is traveling faster at time $t = 80$ seconds? Explain your answer.

Part A

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80) - v(0)}{80 - 0} = \frac{49 - 5}{80} = \frac{11}{20} \text{ ft/sec}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80) - v(0)}{80 - 0} = \frac{49 - 5}{80} = \frac{11}{20} \text{ ft/sec}^2$$

Part B

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

6-2b

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t) dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t) dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$

Part C

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

Let $v_B(t)$ be the velocity of rocket B at time t .

$$v_B(t) = \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$

$$2 = v_B(0) = 6 + C$$

$$v_B(t) = 6\sqrt{t+1} - 4$$

$$v_B(80) = 50 > 49 = v(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.

6-2b



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

Let $v_B(t)$ be the velocity of rocket B at time t .

$$v_B(t) = \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$

$$2 = v_B(0) = 6 + C$$

$$v_B(t) = 6\sqrt{t+1} - 4$$

$$v_B(80) = 50 > 49 = v(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.

6-2b

50.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

(a) Estimate $f'(4)$. Show the work that leads to your answer.

(b) Evaluate $\int_2^{13} (3 - 5f'(x)) dx$. Show the work that leads to your answer.

(c) Use a left Riemann sum with subintervals indicated by the data in the table to approximate $\int_2^{13} f(x) dx$. Show the work that leads to your answer.

(d) Suppose $f'(5) = 3$ and $f''(x) < 0$ for all x in the closed interval $5 \leq x \leq 8$. Use the line tangent to the graph of f at $x = 5$ to show that $f(7) \leq 4$. Use the secant line for the graph of f on $5 \leq x \leq 8$ to show that $f(7) \geq 4/3$.

Part A

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$

Part B

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned} \int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3 dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8 \end{aligned}$$

6-2b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8\end{aligned}$$

Part C

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$

Part D

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

6-2b

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.

6-2b

51.



t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

(a) Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate

$\int_0^{40} v(t) dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of $\int_0^{40} v(t) dt$ in terms of the plane's flight.

(b) Based on the values in the table, what is the smallest number of instances at which the acceleration of the plane could equal zero on the open interval $0 < t < 40$? Justify your answer.

(c) The function f , defined by $f(t) = 6 + \cos\left(\frac{t}{10}\right) + 3 \sin\left(\frac{7t}{40}\right)$, is used to model the velocity of the plane, in miles per minute, for $0 \leq t \leq 40$. According to this model, what is the acceleration of the plane at $t = 23$? Indicate units of measure.

(d) According to the model f , given in part (c), what is the average velocity of the plane, in miles per minute, over the time interval $0 \leq t \leq 40$?

Part A

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$

The integral gives the total distance in miles that the plane flies during the 40 minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is

$10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$

6-2b

The integral gives the total distance in miles that the plane flies during the 40 minutes.

Part B

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval $(0,15)$ and somewhere in the interval $(25, 30)$. Therefore the acceleration will equal 0 for at least two values of t .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval $(0,15)$ and somewhere in the interval $(25, 30)$. Therefore the acceleration will equal 0 for at least two values of t .

Part C

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$

Part D

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

6-2b

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

= 5.916 miles per minute



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

= 5.916 miles per minute

6-2b

52.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	-120	-150

Train A runs back and forth on an east-west section of railroad track. Train A 's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

- (a) Find the average acceleration of train A over the interval $2 \leq t \leq 8$.
- (b) Do the data in the table support the conclusion that train A 's velocity is -100 meters per minute at some time t with $5 < t < 8$? Give a reason for your answer.
- (c) At time $t = 2$, train A 's position is 300 meters east of the Origin Station, and the train is moving to the east. Write an expression involving an integral that gives the position of train A , in meters from the Origin Station, at time $t = 12$. Use a trapezoidal sum with three subintervals indicated by the table to approximate the position of the train at time $t = 12$.
- (d) A second train, train B , travels north from the Origin Station. At time t the velocity of train B is given by $v_B(t) = -5t^2 + 60t + 25$, and at time $t = 2$ the train is 400 meters north of the station. Find the rate, in meters per minute, at which the distance between train A and train B is changing at time $t = 2$.

Part A

1 point is earned for average acceleration

$$\text{average accel} = \frac{v_A(8) - v_A(2)}{8 - 2} = \frac{-120 - 100}{6} = -\frac{110}{3} \text{ m/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for average acceleration

$$\text{average accel} = \frac{v_A(8) - v_A(2)}{8 - 2} = \frac{-120 - 100}{6} = -\frac{110}{3} \text{ m/min}^2$$

Part B

1 point is earned for $v_A(8) < -100 < v_A(5)$

1 point is earned for the conclusion, using IVT

v_A is differentiable $\Rightarrow v_A$ is continuous

6-2b

$$v_A(8) = -120 < -100 < 40 = v_A(5)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $v_A(t) = -100$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $v_A(8) < -100 < v_A(5)$

1 point is earned for the conclusion, using IVT

v_A is differentiable $\Rightarrow v_A$ is continuous

$$v_A(8) = -120 < -100 < 40 = v_A(5)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $v_A(t) = -100$.

Part C

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$\begin{aligned}
 s_A(12) &= s_A(2) + \int_2^{12} v_A(t) dt = 300 + \int_2^{12} v_A(t) dt \\
 \int_2^{12} v_A(t) dt &\approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2} \\
 &= -450 \\
 s_A(12) &\approx 300 - 450 = -150
 \end{aligned}$$

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

6-2b

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$s_A(12) = s_A(2) + \int_2^{12} v_A(t) dt = 300 + \int_2^{12} v_A(t) dt$$

$$\int_2^{12} v_A(t) dt \approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2}$$

$$= -450$$

$$s_A(12) \approx 300 - 450 = -150$$

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.

Part D

2 points are earned for implicit differentiation of distance relationship

1 point is earned for the answer

Let x be train A 's position, y train B 's position, and z the distance between train A and train B .

$$z^2 = x^2 + y^2 \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = 300, y = 400 \Rightarrow z = 500$$

$$v_B(2) = -20 + 120 + 25 = 125$$

$$500 \frac{dz}{dt} = (300)(100) + (400)(125)$$

$$\frac{dz}{dt} = \frac{80000}{500} = 160 \text{ meters per minute}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for implicit differentiation of distance relationship

1 point is earned for the answer

Let x be train A 's position, y train B 's position, and z the distance between train A and train B .

6-2b

$$z^2 = x^2 + y^2 \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = 300, y = 400 \Rightarrow z = 500$$

$$v_B(2) = -20 + 120 + 25 = 125$$

$$500 \frac{dz}{dt} = (300)(100) + (400)(125)$$

$$\frac{dz}{dt} = \frac{80000}{500} = 160 \text{ meters per minute}$$

6-2b

53. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

(a) Use the trapezoidal rule with four equal subdivisions of the closed interval $[0,1]$ to approximate $F(1)$.

(b) On what intervals is F increasing?

(c) If the average rate of change of F on the closed interval $[1,3]$ is k , find $\int_1^3 \sin(t^2) dt$ in terms of k .

Part A

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin \left(\frac{1}{4}\right)^2 + 2 \sin \left(\frac{1}{2}\right)^2 + 2 \sin \left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin \left(\frac{1}{4}\right)^2 + 2 \sin \left(\frac{1}{2}\right)^2 + 2 \sin \left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

6-2b

 ≈ 0.316

Part B

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

1 point is earned for correctly analyzing sign of F' within $[0, 3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

1 point is earned for correctly analyzing sign of F' within $[0, 3]$



1 point is earned for correctly indicating F increases where student's F' is positive

6-2b

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$

Part C

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$



0	1	2
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The student response earns all of the following points:

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$

6-2b

54.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, in gallons per hour, is given by a differentiable function R of time t . The table above shows the rate as measured every 3 hours for a 24-hour period.

- (a) Use a midpoint Riemann sum with 4 subdivisions of equal length to approximate $\int_0^{24} R(t)dt$. Using correct units, explain the meaning of your answer in terms of water flow.
- (b) Is there some time t , $0 < t < 24$, such that $R'(t) = 0$? Justify your answer.
- (c) The rate of water flow $R(t)$ can be approximated by $Q(t) = \frac{1}{79}(768 + 23t - t^2)$. Use $Q(t)$ to approximate the average rate of water flow during the 24-hour time period. Indicate units of measure.

Part A

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

$$\begin{aligned}
 \int_0^{24} R(t)dt &\approx 6[R(3) + R(9) + R(15) + R(21)] \\
 &= 6[10.4 + 11.2 + 11.3 + 10.2] \\
 &= 258.6 \text{ gallons}
 \end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

6-2b

$$\begin{aligned}
 \int_0^{24} R(t) dt &\approx 6 [R(3) + R(9) + R(15) + R(21)] \\
 &= 6 [10.4 + 11.2 + 11.3 + 10.2] \\
 &= 258.6 \text{ gallons}
 \end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

Part B

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t , $0 < t < 24$, such that $R'(t) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t , $0 < t < 24$, such that $R'(t) = 0$.

Part C

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$\begin{aligned}
 &= \frac{1}{24} \int_0^{24} \frac{1}{79} (768 + 23t - t^2) dt \\
 &= 10.785 \text{ gal/hr or } 10.784 \text{ gal/hr}
 \end{aligned}$$

1 point is earned for units

(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

6-2b



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$\begin{aligned} &= \frac{1}{24} \int_0^{24} \frac{1}{79} (768 + 23t - t^2) dt \\ &= 10.785 \text{ gal/hr or } 10.784 \text{ gal/hr} \end{aligned}$$

1 point is earned for units

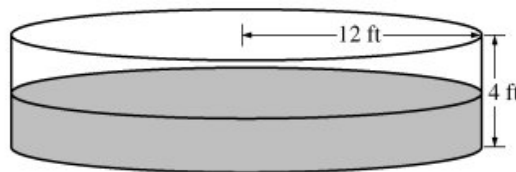
(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

6-2b

55.



t	0	2	4	6	8	10	12
$P(t)$	0	46	53	57	60	62	63



The figure above shows an aboveground swimming pool in the shape of a cylinder with a radius of 12 feet and a height of 4 feet. The pool contains 1000 cubic feet of water at time $t = 0$. During the time interval $0 \leq t \leq 12$ hours, water is pumped into the pool at the rate $P(t)$ cubic feet per hour. The table above gives values of $P(t)$ for selected values of t . During the same time interval, water is leaking from the pool at the rate $R(t)$ cubic feet per hour, where $R(t) = 25e^{-0.05t}$. (Note: The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

- Use a midpoint Riemann sum with three subintervals of equal length to approximate the total amount of water that was pumped into the pool during the time interval $0 \leq t \leq 12$ hours. Show the computations that lead to your answer.
- Calculate the total amount of water that leaked out of the pool during the time interval $0 \leq t \leq 12$ hours.
- Use the results from parts (a) and (b) to approximate the volume of water in the pool at time $t = 12$ hours. Round your answer to the nearest cubic foot.
- Find the rate at which the volume of water in the pool is increasing at time $t = 8$ hours. How fast is the water level in the pool rising at $t = 8$ hours? Indicate units of measure in both answers.

Part A

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} P(t) dt \approx 46 \cdot 4 + 57 \cdot 4 + 62 \cdot 4 = 660 \text{ ft}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for midpoint sum

1 point is earned for the answer

$$\int_0^{12} P(t) dt \approx 46 \cdot 4 + 57 \cdot 4 + 62 \cdot 4 = 660 \text{ ft}^3$$

Part B

1 point is earned for the integral

1 point is earned for the answer

6-2b

$$\int_0^{12} R(t) dt = 225.594 \text{ ft}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{12} R(t) dt = 225.594 \text{ ft}^3$$

Part C

1 point is earned for the answer

$$1000 + \int_0^{12} P(t) dt - \int_0^{12} R(t) dt = 1434.406$$

At time $t = 12$ hours, the volume of water in the pool is approximately 1434 ft^3 .



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$1000 + \int_0^{12} P(t) dt - \int_0^{12} R(t) dt = 1434.406$$

At time $t = 12$ hours, the volume of water in the pool is approximately 1434 ft^3 .

Part D

1 point is earned for $V'(8)$

6-2b

1 point is earned for equation relating $\frac{dV}{dt}$ and $\frac{dh}{dt}$

1 point is earned for $\left. \frac{dh}{dt} \right|_{t=8}$

1 point is earned for units of ft^3/hr and ft/hr

$$V'(t) = P(t) - R(t)$$

$$V'(8) = P(8) - R(8) = 60 - 25e^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{hr}$$

$$V = \pi(12)^2 h$$

$$\frac{dV}{dt} = 144\pi \frac{dh}{dt}$$

$$\left. \frac{dh}{dt} \right|_{t=8} = \frac{1}{144\pi} \cdot \left. \frac{dV}{dt} \right|_{t=8} = 0.095 \text{ or } 0.096 \text{ ft}/\text{hr}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $V'(8)$

1 point is earned for equation relating $\frac{dV}{dt}$ and $\frac{dh}{dt}$

1 point is earned for $\left. \frac{dh}{dt} \right|_{t=8}$

1 point is earned for units of ft^3/hr and ft/hr

$$V'(t) = P(t) - R(t)$$

$$V'(8) = P(8) - R(8) = 60 - 25e^{-0.4} = 43.241 \text{ or } 43.242 \text{ ft}^3/\text{hr}$$

6-2b

$$V = \pi(12)^2 h$$

$$\frac{dV}{dt} = 144\pi \frac{dh}{dt}$$

$$\left. \frac{dh}{dt} \right|_{t=8} = \frac{1}{144\pi} \cdot \left. \frac{dV}{dt} \right|_{t=8} = 0.095 \text{ or } 0.096 \text{ ft/hr}$$

6-2b

56.



t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

(a) Use data from the table to find an approximation for $W'(12)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Approximate the average temperature, in degrees Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using a trapezoidal approximation with subintervals of length $\Delta t = 3$ days.

(c) A student proposes the function P , given by $P(t) = 20 + 10te^{(-t/3)}$, as a model for the temperature of the water in the pond at time t , where t is measured in days and $P(t)$ is measured in degrees Celsius. Find $P'(12)$. Using appropriate units, explain the meaning of your answer in terms of water temperature.

(d) Use the function P defined in part (c) to find the average value, in degrees Celsius, of $P(t)$ over the time interval $0 \leq t \leq 15$ days.

Part A

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

6-2b

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$

Part B

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$

Part C

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^{\circ}\text{C/day} \end{aligned}$$

This means the the temperature is decreasing at the rate of $0.549^{\circ}\text{C/day}$ when $t = 12$ days.

6-2b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^\circ\text{C/day} \end{aligned}$$

This means the the temperature is decreasing at the rate of 0.549°C/day when $t = 12$ days.

Part D

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ\text{C}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ\text{C}$$

6-2b

57.



t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of

$\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

6-2b

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
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The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\begin{aligned} \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

6-2b



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\begin{aligned}\frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79\end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\ &= 71.0 + 2.043155 = 73.043\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\ &= 71.0 + 2.043155 = 73.043\end{aligned}$$

6-2b

58.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

 The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016)$$

6-2b

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned} \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

6-2b



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$